Alpha decay 2

Let ν be the rate of tunneling attempts. Let $e^{-2\gamma}$ be the tunneling probability. Then the decay rate λ is

$$\lambda = \nu e^{-2\gamma}$$

The half-life $\tau_{1/2}$ is

$$\tau_{1/2} = \frac{\log 2}{\lambda} = \frac{\log 2}{\nu}\, e^{2\gamma}$$

Take the log of both sides.

$$\log \tau_{1/2} = 2\gamma - \log \nu + \log(\log 2)$$

Gamow theorized that ν is proportional to r_1. By the approximation $r_1 \propto A^{1/3}$ we have

$$\log \tau_{1/2} = 2\left(C_1 \frac{Z}{\sqrt{E}} - C_2\sqrt{Zr_1}\right) - C_3 \log A + \log(\log 2)$$

where C_3 is estimated to be

$$C_3 = 10.1$$

For the alpha decay process ${}^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th}$ we have

$$\begin{aligned} A &= 238 \\ Z &= 90 \\ E &= 4.27\,\mathrm{MeV} \end{aligned}$$

Hence

$$\log \tau_{1/2} = 39.9$$

and

$$\tau_{1/2} = e^{39.9}\,\mathrm{s} = 2.2 \times 10^{17}\,\mathrm{s} = 7.0 \times 10^{9}\,\mathrm{yr}$$

The actual half-life of ${}^{238}\mathrm{U}$ is

$$4.5 \times 10^{9}\,\mathrm{yr}$$

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